

```
10 iterations: 3.1424259850011: Percent Error: 0.026525762668592
100 iterations: 3.14160098692313: Percent Error: 0.000265258238452891
1000 iterations: 3.14159273692312: Percent Error: 2.65258226508141e-06
10000 iterations: 3.14159265442313: Percent Error: 2.65260522161831e-08
100000 iterations: 3.14159265359816: Percent Error: 2.663891243208e-10
999999999 iterations: 3.1415926535898: Percent Error: 2.40308575932799e-13
```

What factors affect the accuracy of the value of π you calculate?

- The main factor affecting the calculation is the number of iterations N of the Riemann Sum. As N increases, the absolute error decreases and as N approaches infinity, this sum becomes exact.
- There is also computational expense associated with larger and larger N values

What is the most accurate value of π you can get, and what is the corresponding value of N ?

- 3.1415926535898 is the value of π I calculated.
- This is with a percent error of $2.4\text{e-}13\%$ compared to `<cmath>` π value, which was after 999,999,999 iterations. This error is so small that it is approaching machine precision for doubles.
- In theory, you can run this program for a larger value of N and reach machine precision, but this is computationally very expensive.
- For example, you could use `long` types to get larger values of N , but at this point we reach a point of diminishing returns.